

REAL-WORLD ASSET TOKENISATION

A Game Changer for Global Trade

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A Comprehensive Analysis of DLT-Enabled Trade Finance and the Tokenisation of Real-World Assets

RESEARCH & ANALYSIS

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Executive Summary

The global economy stands at an inflection point. After decades of incremental digitalisation, a more fundamental transformation is now underway, one that reaches into the very DNA of how value is created, transferred and held. Real-world asset tokenisation, the process of converting ownership rights in tangible and financial assets into programmable digital tokens on distributed ledgers, is rapidly moving from theoretical promise to institutional practice.

This paper makes a straightforward argument: tokenisation is not merely a fintech novelty. It is the most significant architectural upgrade to global financial infrastructure since the invention of the joint-stock company. And nowhere is its potential more consequential, or more urgently needed, than in global trade finance.

\$32B+

On-chain RWA market value, May 2026

\$2.5T

Global trade finance gap, 2023 (ADB)

The numbers are striking. The on-chain real-world asset market (excluding stablecoins) surpassed \$32 billion in distributed value in May 2026, representing growth of over 380% in just three years. Projections from major financial institutions and consultancies suggest this figure could reach \$30 trillion or beyond by the mid-2030s. Against that backdrop, a \$2.5 trillion trade finance gap, the chasm between the financing that global SMEs need and what they actually receive, represents both a humanitarian failure and a market opportunity of historic proportions.

Tokenisation offers a credible bridge to this opportunity. By converting trade finance assets, letters of credit, supply chain receivables, and working capital loans into transferable digital instruments, tokenisation can bring transparency, liquidity, and accessibility to an asset class that has historically been locked inside the balance sheets of a small number of global banks. It can lower the cost of capital for the 'missing middle' of mid-market enterprises in Asia, Africa and the Middle East. It can enable institutional investors to access short-tenor, self-liquidating, low-correlation assets with strong risk-adjusted returns. And it can help central banks, multilateral institutions and regulators achieve the financial inclusion objectives that decades of conventional policy have failed to meet.

This paper synthesises the evidence, analyses the mechanics, maps the emerging institutional landscape and confronts the real risks honestly. It concludes with a call to action for investors, banks, regulators and policymakers who recognise that the window of first-mover advantage is narrowing.

“The question is no longer whether tokenisation will transform global finance. The question is who will lead the transformation and who will be left behind.”

Methodology and Scope

RWA market size: For the purposes of this paper, the term “RWA market” refers to tokenised real-world assets recorded and tracked by market-data providers, excluding stablecoins unless otherwise specified. Market estimates vary across sources because definitions differ regarding the inclusion of tokenised treasuries, private credit, funds, commodities and other asset classes.

Tokenised trade finance: In this paper, “tokenised trade finance” refers to trade receivables, letters of credit, supply-chain finance assets or related claims that are represented as tokens on a distributed ledger and linked to legally enforceable off-chain rights. Pure payment tokens, generic crypto-assets and unaffiliated decentralised finance lending are excluded unless they are directly tied to trade finance assets.

Trade finance gap: The paper adopts the Asian Development Bank’s definition of the trade finance gap as unmet demand for trade finance, measured as the difference between financing requested by firms and financing approved or supplied by financial institutions.

Summary footnote

Methodology note: This paper uses “RWA market” to mean tokenised real-world assets tracked by market-data providers, excluding stablecoins unless otherwise stated. “Tokenised trade finance” refers to trade receivables, letters of credit and related claims represented on a distributed ledger and linked to enforceable off-chain rights. The “trade finance gap” follows the Asian Development Bank’s definition of unmet demand, measured as the difference between requested and supplied trade finance.

CHAPTER 1

The Architecture of Value: What Is Asset Tokenisation?

Before examining the consequences of tokenisation, we must understand precisely what it is, and, equally importantly, what it is not. In a decade awash with breathless technology claims, definitional rigour is not mere pedantry. It is the foundation of sound analysis.

The Core Definition

Asset tokenisation is the process of creating a digital representation of a real or traditional asset on a distributed ledger technology (DLT) platform, typically a blockchain. The resulting token is a programmable, cryptographically secured certificate of ownership, or of a specific right related to ownership, that can be transferred, subdivided, collateralised and traded with a speed, efficiency and global reach that traditional paper-based or centralised-ledger systems cannot match.

The definition encompasses a vast spectrum of assets. Physical assets – real estate, commodities, artworks, and infrastructure, can be tokenised. Financial assets – bonds, equities, loans, and trade receivables, can also be tokenised. Even future cash flows, intellectual property rights and carbon credits can be represented on-chain. The common thread is the conversion of analogue ownership into programmable digital form.

“A token is not merely a digital record of an asset. It is the asset re-imagined as code, programmable, composable and globally accessible.”

The Anatomy of a Token

Understanding tokenisation requires understanding what a token actually contains. A well-structured token for a financial asset typically encodes the following:

- Identity and provenance: Who originated the asset, when, and under what legal structure.
- Economic rights: What cashflows, dividends, interest payments or redemption values attach to the token?
- Governance rules: What decisions token holders can influence or vote on.
- Compliance metadata: KYC/AML status of eligible holders, jurisdictional restrictions, and accreditation requirements.
- Transfer logic: Under what conditions the token can be transferred, subdivided or redeemed.

- Oracle links: Real-time connections to off-chain data sources that trigger on-chain actions, for example, releasing payment upon confirmed delivery of goods.

This architecture makes the token fundamentally different from a traditional security or loan agreement. Rules do not merely constrain the asset from outside; they are embedded within it. The asset becomes self-enforcing within the parameters set by its creator and the governing legal framework.

Fungible vs. Non-Fungible Tokens

A critical technical distinction determines the appropriate token structure for different asset types. Fungible tokens are interchangeable; one unit is identical to every other unit of the same class. ERC-20 tokens on the Ethereum blockchain are the canonical example. Currency, money market shares and standardised receivables are natural candidates for fungible tokenisation.

Non-fungible tokens (NFTs) are unique; each carries distinct attributes. This structure is appropriate for representing individual assets: a specific trade receivable with a particular counterparty, maturity date and principal amount; a specific piece of real estate with its unique legal title and physical characteristics; and a specific artwork with its provenance and condition.

In sophisticated trade finance tokenisation, both types work in concert: individual receivables are tokenised as NFTs to preserve their unique attributes, and then pooled into tranches represented by fungible tokens for distribution to investors. This architecture, pioneered in several landmark pilot programmes, enables institutional-scale investment in what would otherwise be impractically granular assets.

Smart Contracts: The Engine of Automation

No discussion of tokenisation is complete without examining smart contracts, the self-executing programmes that give tokenised assets their most powerful properties. A smart contract is code deployed on a blockchain that automatically executes predefined actions when specified conditions are met, without requiring human intermediation.

In trade finance, the implications are profound. A smart contract can automatically release payment to a supplier when a shipment's location data confirms delivery at the destination port. It can automatically redirect interest payments to a new holder when a token is transferred. It can automatically trigger a default notification and initiate collateral liquidation when a borrower fails to make a scheduled payment. It can automatically adjust the outstanding balance of a revolving credit facility as new receivables are added to the pool.

Each of these actions, in traditional trade finance, requires manual intervention by multiple parties, document checkers, compliance officers, relationship managers, and clearing agents, at every step. In specific trade finance workflows, smart contracts can materially reduce processing time and create a more transparent audit trail, although they do not eliminate the legal and operational steps involved in the full transaction lifecycle.

Public vs. Permissioned Blockchains

A final architectural choice that significantly shapes tokenisation outcomes is the selection of the underlying blockchain infrastructure. Public blockchains, Ethereum being the most prominent, are open to anyone to read, write and participate in. They offer maximum transparency and composability (tokens can interact with any other protocol on the same chain) but raise legitimate concerns about privacy, regulatory compliance and throughput capacity.

Permissioned (or private) blockchains, such as those built on Hyperledger Fabric or R3 Corda, restrict participation to vetted entities. They offer greater control, privacy and regulatory compatibility but sacrifice some of the openness and composability that make public chains powerful.

The financial industry's emerging consensus is a hybrid model: permissioned access layered over public blockchain infrastructure, preserving the trustless settlement finality of public chains while enforcing the identity and compliance requirements that institutional participants demand. This architecture, sometimes called a 'regulated public chain', is increasingly seen as the viable path for scaling tokenised financial markets.

CHAPTER 2

From REITs to the Blockchain: A Historical Arc

Tokenisation did not emerge from a vacuum. Its intellectual lineage stretches back decades, through a series of innovations in financial architecture that progressively democratised access to assets previously available only to the wealthy and the well-connected. Understanding this lineage illuminates both how far we have come and how much further the current wave of innovation can travel.

The Fractionalisation Precursors: 1960s–1990s

The conceptual ancestor of tokenisation is fractionalisation, the idea that a large, illiquid asset can be divided into smaller, more accessible units without losing its fundamental economic character. Real estate investment trusts (REITs), invented in the United States in 1960, were among the first mainstream implementations of this principle. By pooling properties into a corporate vehicle and issuing shares, REITs allowed ordinary investors to participate in commercial real estate markets previously accessible only to institutions and the wealthy.

Exchange-traded funds (ETFs), which emerged in their modern form in the early 1990s, extended this principle to diversified portfolios of securities. By creating redeemable shares that tracked underlying baskets of assets and traded on exchanges throughout the day, ETFs combined the diversification of mutual funds with the liquidity of individual stocks. By the mid-2000s, the ETF market had transformed retail investing globally.

Both REITs and ETFs achieved fractionalisation through centralised intermediary structures, regulated corporate vehicles, exchange listing requirements, custodians and transfer agents. They required, and depended upon, a dense ecosystem of trusted institutions. This created efficiency within established markets but could not extend their benefits to asset classes or geographies where that institutional infrastructure did not exist.

The Blockchain Breakthrough: 2009

The publication of the Bitcoin white paper in October 2008 and the launch of the Bitcoin network in January 2009 represented a conceptual breakthrough of the first order. For the first time, it was possible to transfer value digitally between parties who did not know or trust each other, without relying on any central intermediary, in a manner that was cryptographically verifiable and publicly auditable.

Bitcoin's elegant solution to the 'double-spend problem', how to prevent the same digital asset from being spent twice, opened the door to a new way of thinking about ownership and transfer. If Bitcoin could function as digital money without a central bank, what else could be represented on a distributed ledger?

Ethereum and Programmable Value: 2015

The answer, elaborated by Vitalik Buterin and the Ethereum team, was almost anything. Ethereum, launched in 2015, introduced two features that transformed the blockchain landscape. The Ethereum Virtual Machine enabled programmable logic to run on-chain, which is the smart contracts described in Chapter 1. And the ERC-20 token standard provided a template for creating new tokens on the Ethereum network, with the infrastructure of wallets, exchanges and developer tools immediately available.

The first wave of Ethereum-based tokenisation was, in retrospect, a period of experimentation as much as it was of excess. Initial Coin Offerings (ICOs) in 2017-18 raised billions for projects of widely varying quality. The speculative frenzy obscured genuine technical progress but also stress-tested the infrastructure and generated a global community of developers, investors and observers.

Security Token Offerings and Regulatory Engagement: 2018–2020

The late-2010s discussion of security tokens and compliant token offerings marked an important regulatory turning point, but it would be more accurate to say that the SEC helped clarify how tokenised instruments could fit within existing securities law rather than ‘articulated the concept of STOs.

In the same period, major financial institutions began investing in DLT infrastructure. JPMorgan launched the Interbank Information Network (later Liink) for cross-border payments. HSBC, Deutsche Bank and Standard Chartered experimented with trade finance platforms. The Bank for International Settlements established the BIS Innovation Hub to coordinate central bank experimentation with digital finance. These were not yet tokenisation at scale, but they were the foundation-laying that would make scale possible.

The Real-World Asset Moment: 2020 to Present

The COVID-19 pandemic paradoxically accelerated tokenisation by exposing the fragility of analogue financial infrastructure. Paper-based trade finance processes ground to a halt when physical documents could not be signed or couriered. Letters of credit that relied on wet signatures became unworkable. The case for digital alternatives became urgent rather than theoretical.

From 2020 onwards, the tokenisation of real-world assets emerged as a distinct and serious market segment. Several developments defined this period. Regulatory frameworks began to crystallise in forward-looking jurisdictions – Singapore, Switzerland, the UAE, and the UK – providing the legal clarity that institutional participants needed. Major asset managers moved beyond observation to participation: BlackRock's BUIDL fund, launched in 2024, became the world's largest tokenised fund. Franklin Templeton deployed on-chain money market funds. JPMorgan's Kinexys platform processed trillions in intraday repo and settlement transactions.

By May 2026, on-chain real-world assets excluding stablecoins had reached a multi-tens-of-billions scale, according to market-data providers, reflecting rapid growth over the prior three years. Tokenisation has moved beyond experimentation and into an emerging

market phase, but it is still modest in scale compared with traditional capital markets and remains fragmented across asset classes and platforms.”

Milestone	Year	Significance
REITs introduced (US)	1960	First mainstream fractionalisation of real assets
ETFs launched (US)	1993	Fractionalisation extended to diversified portfolios
Bitcoin whitepaper	2008	Trustless digital value transfer without intermediaries
Ethereum launch	2015	Smart contracts enable programmable tokenisation
SEC coins 'STO'	2018	Regulatory framework for compliant token issuance
COVID-19 paper crisis	2020	Urgency for digital trade finance infrastructure
First RWA tokenisation pilots	2022	Institutional-scale proof of concept
BlackRock BUIDL fund	2024	World's largest tokenised fund launched
RWA market > \$32 billion	2026	Mainstream institutional adoption underway

CHAPTER 3

The Trade Finance Crisis: A \$2.5 Trillion Problem

To appreciate why trade finance tokenisation matters, it is necessary to first understand the scale and nature of the problem it addresses. The global trade finance gap is not a technical inconvenience. It is a structural failure of the financial system that costs developing economies billions in lost output, condemns millions of small businesses to suboptimal terms or outright exclusion, and leaves a multi-trillion dollar investment opportunity effectively off the table for institutional capital.

The Scale of the Gap

\$2.5 Trillion

Global Trade Finance Gap — Asian Development Bank, 2023

According to the Asian Development Bank's 2023 Trade Finance Gaps, Growth, and Jobs Survey, the global trade finance gap, the difference between the financing that businesses request and what they actually receive, reached \$2.5 trillion. This represents a 47% increase from \$1.7 trillion in 2020, the largest single-period increase since the metric was introduced. The COVID-19 pandemic, geopolitical instability, rising interest rates and increased risk-aversion among traditional lenders all contributed to this dramatic widening.

To contextualise the magnitude, the trade finance gap represents approximately 10% of total global trade exports. Since conventional trade finance currently covers around 80% of all exports, the gap is not a marginal inefficiency; it is a structural exclusion of a significant segment of global commercial activity.

There is also reason to believe the true shortfall is substantially larger. Businesses rejected for trade finance, or that have learnt not to apply because rejection is expected, do not appear in the ADB's denominator. When the International Finance Corporation estimates that 65 million firms, 40% of all formal micro, small and medium enterprises in developing countries, have unmet financing needs, it suggests that the documented \$2.5 trillion gap is only the visible portion of a much larger iceberg. Total true demand, including suppressed demand from enterprises that do not apply, may exceed \$5 trillion.

Who Bears the Burden?

Trade finance is not withheld uniformly. The gap falls with crushing disproportionality on specific types of businesses and geographies.

Geographically, the burden falls heaviest on developing and emerging market economies in Asia, Africa and the Middle East, precisely the regions where trade growth is fastest and where the multiplier effects of trade on employment and development are most significant. Sub-Saharan Africa faces an SME financing gap alone estimated at over \$331 billion. Cross-border blockchain-based payment rails may reduce certain transaction costs and processing frictions compared with conventional correspondent banking, although the magnitude of the savings varies widely by corridor, payment type and operating model. Their potential is significant, but it should be described as conditional rather than universal.

By business size, the exclusion intensifies as company size decreases. Large, investment-grade corporations have ready access to trade finance at competitive rates, their creditworthiness is well understood, their documentation is standardised, and their relationship banks are eager to serve them. Microfinance institutions and development finance programmes pay some attention to small and micro enterprises at the retail end of the spectrum. But the 'missing middle' – what this analysis calls the 'middle market enterprises' (MMEs) – falls through the gap between these two segments.

“Middle-market enterprises: too large for microfinance, too opaque for investment-grade lending, and too important to the global economy to ignore.”

The Missing Middle

Middle-market enterprises typically have annual revenues between \$10 million and \$500 million. They are active across all sectors but are disproportionately represented in manufacturing, agriculture, and natural resources, which are sectors with significant cross-border trade activity. They are particularly prevalent in the fast-growing economies of South and Southeast Asia, sub-Saharan Africa, the Gulf Cooperation Council states and Latin America.

What makes MMEs especially difficult to finance through conventional channels is not their creditworthiness; their actual default rates, where data exist, are often comparable to or better than those of investment-grade corporates. The problem is informational. MMEs lack the audited financial statements, credit ratings, legal documentation and institutional relationships that enable banks to efficiently assess and price their risk. The operational cost of credit assessment for an MME is often not materially lower than for a large corporate, but the margin is far smaller and the volume of individual transactions far lower. The economics of conventional trade finance simply do not work for this segment at scale.

The Structural Deficiencies of Conventional Trade Finance

Even where trade finance is available, the process through which it is delivered is an anachronism. Consider the life cycle of a conventional letter of credit, still the dominant

instrument for international trade in many markets. A letter of credit is initiated on paper, transmitted by fax or courier, verified manually by document checkers at multiple banks, physically archived in multiple jurisdictions, and settled through a chain of correspondent banking relationships that may involve five or more institutions, which can take 30 to 90 days to process. Each step introduces the possibility of error, dispute or delay. Each intermediary extracts a fee.

The human capital requirements are substantial. Trade finance compliance, anti-money laundering checks, sanctions screening, know-your-customer verification, and documentary compliance review require skilled specialists whose cost is increasingly difficult to justify for small-value transactions. Many banks have simply exited the trade finance business in developing market corridors, calculating that the compliance burden makes the economics unworkable.

The result is a market characterised by high barriers, low transparency, limited secondary market liquidity, and a structural bias toward large, well-known counterparties. These are precisely the conditions that tokenisation is best positioned to address.

The Investment Opportunity Hidden in the Gap

From the perspective of institutional investors, trade finance assets possess attributes that are genuinely rare in contemporary capital markets: short tenors (typically 30–180 days), low correlation with equity and bond markets, strong recovery rates in default scenarios, and direct linkage to real economic activity rather than financial engineering.

Asset Type	Default Rate	Recovery Rate
Import Letter of Credit	0.40%	62.7%
Export Letter of Credit	0.04%	63.7%
Loans for Import/Export	0.75%	62.3%
Performance Guarantee	0.47%	98.6%
US High-Yield Bond	3.40%	37.3%
US Leveraged Loan	3.66%	59.4%

Source: Allianz Global Investors data. The comparison is striking: trade finance instruments, across product types, exhibit default rates approximately an order of magnitude lower than public credit instruments, while recovery rates are broadly comparable or superior. For institutional investors seeking yield, diversification and stability, trade finance should be a highly attractive asset class.

Yet institutional investment in trade finance has historically been minimal, constrained by a lack of standardisation, the operational complexity of the asset class, insufficient transparency about underlying assets, and the absence of liquid secondary markets. Tokenisation directly addresses each of these barriers.

CHAPTER 4

The Mechanics of Tokenised Trade Finance

Understanding the mechanics of trade finance tokenisation, how an underlying receivable or loan actually becomes a tradeable digital token, is essential for evaluating both its potential and its limitations. The process is more complex than simply 'putting an asset on the blockchain', and the design choices made at each stage have significant implications for investor protection, legal enforceability, liquidity and operational efficiency.

Step One: Asset Origination and Selection

The tokenisation process begins with the identification and selection of appropriate underlying assets. Not all trade finance assets are equally suitable candidates. The most tokenisation-friendly assets share several characteristics: they are self-liquidating (generating their own repayment through the commercial transaction they finance), they have defined maturities, they are legally assignable without the obligor's consent, and they are accompanied by sufficient documentary evidence to enable due diligence.

Letters of credit, trade receivables, invoice discounting facilities and supply chain finance instruments are generally well-suited. Project finance and commodity loans with long tenors and complex covenant structures present greater challenges. The art of trade finance tokenisation lies partly in asset selection, identifying the subset of the trade finance universe that meets the transparency, assignability and documentary standards that investors and regulators will require.

Step Two: Legal Structuring and True Sale

The legal architecture underlying the tokenisation is, in many respects, more important than the technology. A token that does not represent a legally enforceable claim on an underlying asset is worthless, or worse, fraudulent. The foundational legal requirement is a 'true sale' of the underlying assets from the originating bank to a special purpose vehicle (SPV), which then issues tokens that are backed by those assets.

A true sale, as distinct from a secured loan, removes the assets from the originator's balance sheet and protects them from the originator's insolvency. It creates bankruptcy remoteness: investors who hold tokens backed by the SPV's assets should, in principle, be unaffected by financial difficulties at the originating bank. Establishing true sale treatment across multiple jurisdictions, particularly in emerging markets where legal frameworks for asset-backed securities are less developed, is among the most challenging and consequential aspects of international trade finance tokenisation.

The legal structure must also address questions of governing law, jurisdiction for dispute resolution, and the legal status of the token itself (is it a security, a note, or an instrument?), and the mechanism by which token holders can enforce their rights against the underlying obligors in the event of default.

Step Three: Risk Tranching

Most sophisticated tokenised trade finance structures involve tranching, which allocates assets to different risk and return profiles for different investor segments. In a typical structure, underlying receivables (perhaps tokenised individually as NFTs) are pooled and then allocated to senior and junior tranches represented by fungible tokens.

Senior token holders have priority claims on cashflows and are exposed to loss only after the junior tranche is fully impaired. They receive a lower but more predictable return. Junior token holders absorb first losses and receive residual cashflows, effectively an excess spread above the senior coupon. This structure allows the same pool of assets to serve investors with different risk appetites: conservative institutional investors such as insurance companies or pension funds in the senior tranche and higher-yield-seeking investors in the junior tranche.

The granularity of the tranching structure is one of the key advantages of tokenisation over conventional securitisation. Whereas traditional ABS structures are typically binary or have a small number of tranches, smart-contract-based tokenisation can theoretically create continuous risk/return curves with near-unlimited granularity, enabling precise matching of investor preferences with specific risk exposures.

Step Four: Smart Contract Programming

Once the legal structure is established and the risk tranching is determined, the economic terms of the tokens are encoded in smart contracts on the chosen blockchain. The smart contracts automate the lifecycle management of the tokens, which includes distributing interest payments, repaying principal upon asset maturity, adding new receivables to replenish the pool as existing ones mature, and handling defaults and recovery distributions.

Smart contract programming for financial assets requires extraordinary care. Bugs in smart contract code have led to the permanent loss of hundreds of millions of dollars in the broader crypto market. For institutional-grade financial products, rigorous auditing of smart contract code by multiple independent firms with relevant expertise is non-negotiable. The irreversibility of blockchain transactions means that errors cannot be corrected retroactively; prevention is the only viable strategy.

“In a blockchain-based system, the code is the contract. If the code is wrong, so is the contract, and there may be no remedy.”

Step Five: Token Issuance and Distribution

Tokens are typically issued through an Initial Token Offering (ITO), the functional equivalent of an IPO or a bond issuance in traditional capital markets. The ITO may be conducted on the originating bank's proprietary platform, on a regulated third-party tokenisation platform, or, in more ambitious structures, on a public blockchain that is accessible to a global investor base.

Investor eligibility criteria are typically encoded directly in the token's smart contract: only wallets that have completed KYC/AML verification and, where required, demonstrated

accredited investor status can receive tokens. These restrictions travel with the token through secondary market trading, eliminating the compliance risks associated with unrestricted token transfers.

Step Six: Secondary Market Trading and Settlement

One of the most significant advantages of tokenisation over conventional trade finance investment, where secondary market liquidity is nearly non-existent, is the potential for liquid secondary markets. Because tokens are standardised, divisible and transferable on blockchain infrastructure, they can in principle be traded continuously on digital asset exchanges, peer-to-peer or via regulated electronic trading venues.

Settlement is near-instantaneous, typically within seconds on modern blockchain platforms, compared to the T+2 or longer settlement cycles of conventional securities markets. This dramatically reduces counterparty risk and frees up capital that would otherwise be tied up in settlement float. For large institutional investors managing dozens or hundreds of positions simultaneously, this efficiency gain is significant.

The development of secondary markets for tokenised trade finance is still in its early stages. While market infrastructure is expanding and regulated venues are beginning to support tokenised assets, liquidity is still limited; any future improvement will depend on broader adoption, standardisation and investor participation.

CHAPTER 5

Four Pillars of Transformation

The transformative potential of trade finance tokenisation cannot be reduced to a single benefit. It operates simultaneously across four distinct dimensions, each significant in its own right and mutually reinforcing the others. Together, they constitute a qualitative change in the nature of trade finance, not merely an incremental improvement in existing processes but a structural reimagining of who can participate, how transactions are conducted, what information is available and how value is transferred.

Pillar One: Improving Market Access

The first and most immediately legible transformation is the expansion of market access, both for investors seeking exposure to trade finance assets and for businesses seeking financing.

From the investor side, trade finance has historically been the exclusive domain of global commercial banks. These institutions originate trade finance assets for their own portfolios, partly because the information asymmetry between originators and external investors makes arm's-length investment impractical and partly because the operational complexity of the asset class creates high barriers to entry. Institutional investors, asset managers, insurance companies, pension funds, and sovereign wealth funds have had limited ability to access trade finance directly, despite the asset class's attractive risk/return profile.

Tokenisation changes the game by converting trade finance assets into standardised, transferable instruments with transparent documentation and programmable cashflows. The information that was previously locked inside banks, asset quality data, obligor creditworthiness, collateral details, cashflow history, can be encoded into or linked from the token, giving external investors the visibility they need to underwrite exposure independently. Combined with the divisibility of tokens, this opens the asset class to institutional investors who previously lacked the scale or operational infrastructure to participate.

From the borrower side, the expansion of the investor base creates competitive pressure on pricing and availability. When a single SME receivable can be held by hundreds of investors across dozens of jurisdictions, rather than sitting on a single bank's balance sheet, the pool of available capital deepens dramatically, and the cost of capital reflects global risk appetite rather than a single bank's credit committee.

Deep-Tier Supply Chain Finance

A particularly powerful application of tokenisation-enabled market access is deep-tier supply chain finance, extending financing not just to tier-one suppliers of large anchor buyers, but through the supply chain to tier-two, tier-three and beyond. Conventional supply chain finance reaches primarily tier-one suppliers, who have direct contractual relationships with the anchor

buyer. Suppliers further down the chain, often smaller, more numerous, and operating in more challenging markets, are effectively invisible to lenders.

Digital trade tokens, backed by the credit of the anchor buyer and transferable through the supply chain using smart contract logic, change this equation. A tier-two supplier can receive a token from its tier-one customer, use that token as collateral to obtain financing from its local bank, or transfer the token to its own tier-three supplier in payment. The token's value derives from the anchor buyer's creditworthiness, a major corporation with a well-understood credit profile, rather than the small supplier's own financial history. This is the essence of deep-tier supply chain finance: extending institutional-quality credit enhancement all the way to the smallest participants in global trade.

Pillar Two: Simplifying Trade Complexity

Trade finance is operationally complex because international trade is also operationally complex. Multiple parties, jurisdictions, currencies, regulatory regimes, and document types introduce friction, delay, and the possibility of error or fraud. Tokenisation does not eliminate this complexity, but it provides tools to manage it far more efficiently.

The most transformative tool is programmability. Smart contracts can encode the conditions under which various actions occur, payment release, ownership transfer, and collateral liquidation, and execute those actions automatically and verifiably when conditions are met. This removes the human intermediation that is simultaneously the most expensive and most error-prone element of conventional trade finance processing.

Consider the letter of credit: an instrument designed to bridge the trust gap between buyers and sellers in different countries by substituting the creditworthiness of banks for the creditworthiness of the trading parties. In its traditional form, a letter of credit requires the presentation and verification of a suite of physical documents, bill of lading, commercial invoice, packing list, certificate of origin, and insurance certificate, each of which must conform precisely to the terms specified in the credit. Discrepancies, even trivial ones, can delay payment or result in documents being rejected and resubmitted, a process that can take weeks.

A tokenised, smart-contract-based equivalent encodes the documentary requirements digitally. Electronic bills of lading (eBLs), digital invoices, IoT-verified shipping confirmations and customs data can all be linked to the token and verified programmatically. Payment is triggered automatically when the verified data meets the encoded conditions. Discrepancy rates fall dramatically. Processing times shrink from weeks to hours. And the complete transaction record, immutably stored on-chain, provides a definitive audit trail for regulatory, tax and dispute resolution purposes.

“What once took weeks of manual verification and physical document handling can now be completed in hours, automatically, transparently, with an immutable audit trail.”

Pillar Three: Digitalising Securitisation

Traditional securitisation, the bundling of loans or receivables into asset-backed securities for sale to investors, has a long history in financial markets, dating back to the mortgage-backed securities of the 1970s. It is a powerful tool for capital recycling and risk distribution. But conventional securitisation is expensive, slow and operationally intensive. A typical asset-backed securities transaction involves six to twelve months of legal and structuring work, millions in fees to rating agencies, lawyers and investment banks, and results in a relatively illiquid instrument held by a small number of large institutions.

Tokenisation reimagines securitisation for the digital age. Smart contracts automate many of the functions that require human intermediation in conventional structures: the verification of asset eligibility, the calculation and distribution of cashflows, the monitoring of pool performance, and the addition of new assets as existing ones mature. Legal documentation, rather than existing only as paper contracts filed in multiple jurisdictions, is encoded in and enforced by the smart contract.

Critically, tokenisation expands the universe of assets that can be efficiently securitised. Conventional securitisation is economic only above a certain pool size, typically hundreds of millions of dollars, because the fixed costs are so high. A tokenised structure that automates most of these functions changes the economics dramatically, making it viable to securitise smaller pools of assets, shorter-tenor instruments, and asset classes from markets where conventional securitisation infrastructure does not exist.

This is not merely theoretical. Several landmark transactions have demonstrated that tokenised securitisation is operationally viable at a meaningful scale. The next phase of development, moving from bespoke pilot transactions to repeatable, standardised processes at scale, is now underway.

Pillar Four: Reducing Information Asymmetry

Perhaps the most fundamental and least discussed advantage of tokenisation is its potential to reduce information asymmetry in financial markets, which is the gap between what issuers know about their assets and what investors know about them.

Information asymmetry is the root cause of most market failures in finance. It is why banks charge high spreads on SME loans; they know less about the borrower than the borrower knows about themselves, and they must be compensated for that uncertainty. It is why trade finance assets are illiquid in secondary markets; potential buyers lack the information to assess what they are buying. It is why financial crises periodically destroy value on a massive scale; risk is mispriced because those bearing it do not fully understand it.

Tokenisation provides multiple mechanisms for reducing information asymmetry. First, the immutable on-chain record of an asset's origin, terms, payment history and current status is accessible to all authorised participants simultaneously; there is no information advantage available to insiders that is not available to outsiders. Second, the linkage of tokens to real-world data via oracles, IoT sensors, customs databases, payment systems, and shipping platforms provides continuous, verifiable updates on the status of the underlying assets. Third, the

development of listing frameworks for tokenised assets, requiring public disclosure of offering documents, asset quality reports and performance data, creates standardised information disclosure that institutional investors require for due diligence.

Over time, as tokenised asset markets develop transaction histories, the accumulation of on-chain data will enable increasingly sophisticated risk analysis. Machine learning models trained on the detailed, granular, verified data generated by tokenised trade finance transactions will be able to price risk more accurately than any model based on the patchy, delayed and often unreliable data available in conventional markets. This will further reduce the cost of capital for borrowers, improve returns for investors, and deepen market liquidity.

Digitisation Versus Tokenisation

A more balanced assessment of trade finance innovation requires distinguishing between three related but distinct layers of change: document digitisation, system interoperability and asset tokenisation. In many cases, the most immediate gains in trade finance do not come from tokenisation at all, but from simpler forms of digitisation that remove paper, reduce manual error and shorten processing time. Electronic bills of lading, digital trade documents, identity solutions and workflow automation can already deliver meaningful efficiency gains in shipping, compliance and document handling.

A second layer of value comes from interoperability. The trade-finance ecosystem remains fragmented across banks, platforms, jurisdictions and document standards, creating costly “digital islands” even when individual systems are well designed. API-based connectivity and common data standards can link these systems, improve cross-platform data exchange and reduce reconciliation friction without changing the underlying legal structure of the asset itself. In other words, standardisation and connectivity can address a significant share of the trade-finance bottleneck, rather than tokenisation.

Tokenisation becomes most compelling when the objective is not merely to digitise a process, but to transform a trade claim into a transferable financial instrument with programmable rights and obligations. This is where tokenisation adds something qualitatively different: it can combine legal claims, automated compliance, fractional ownership and secondary transferability in a single instrument. That distinction matters because tokenisation is not a universal substitute for digitisation; it is a more specific solution for cases in which the asset itself, rather than only the workflow around it, must be represented, distributed or settled on-chain.

This distinction also helps define the limits of the tokenisation thesis. If eBL adoption, digital identity, or interoperable APIs can solve a trade finance pain point, then tokenisation may add unnecessary complexity. But if the goal is to create a legally enforceable, programmable and transferable representation of receivables or trade claims that can be distributed to multiple investors, then tokenisation provides capabilities that simpler digitisation cannot replicate. The analytical task, therefore, is not to ask whether tokenisation is “better” in the abstract, but to identify where it adds unique value beyond the existing digital trade stack.

Unit Economics

A credible assessment of tokenised trade finance must examine unit economics, not just technological possibility. The central question is whether tokenisation reduces the all-in cost of originating, structuring, servicing and distributing trade assets enough to justify the additional complexity of a token-based architecture. In many cases, the answer will depend less on the blockchain itself than on the surrounding legal, operational and compliance stack.

Origination costs include borrower onboarding, credit assessment, document verification and counterparty due diligence. These are often the most expensive steps in trade finance, particularly for smaller or less transparent borrowers, because the fixed cost of risk assessment does not fall proportionally with transaction size. Tokenisation may reduce some of this friction by standardising data and improving visibility, but it does not eliminate the need for underwriting, legal review or counterparty screening.

Servicing costs cover cashflow administration, payment reconciliation, reporting, investor communications and default management. Here, tokenisation can create meaningful efficiencies if smart contracts automate routine actions and if data feeds reliably update the status of underlying assets. However, those savings are only realised when the asset structure, legal documentation and oracle architecture are sufficiently robust to support automation without introducing new operational risk.

Legal setup costs can be substantial. A tokenised trade finance structure typically requires legal opinions, jurisdictional analysis, true-sale review, SPV structuring, offering documentation and regulatory engagement. These costs are likely to be material in the early stages of market development, especially where the legal treatment of tokenised claims is not yet settled. In practice, the legal cost burden may offset some of the operational savings that tokenisation promises, at least until structures become more standardised.

Custody and infrastructure costs are another important component. Institutional-grade custody of tokenised assets requires secure key management, compliance controls, transfer restrictions and integration with portfolio and risk systems. In addition, issuers and platforms may face ongoing technology, audit, network and interoperability expenses. For tokenisation to scale economically, either standardisation and competition must reduce these costs or sufficiently high transaction volumes must absorb them.

The expected fee structure should also be made explicit. Tokenised trade finance may introduce or redistribute fees across origination, platform access, custody, transaction execution, compliance monitoring and secondary-market servicing. The analytical issue is not whether tokenisation removes fees altogether but whether it lowers total cost, improves transparency and shifts value toward activities that genuinely add economic utility.

CHAPTER 6

The Institutional Landscape: Who Is Building What?

Tokenisation of real-world assets is no longer a fringe preoccupation of crypto-native firms. The landscape of active participants now encompasses the world's largest asset managers, systemically important banks, central banks, multilateral institutions and regulatory bodies. Understanding who is building what, and why, is essential for assessing the credibility and trajectory of the market.

Global Banks: From Observation to Infrastructure

The most significant shift in the tokenisation landscape over the past three years has been the move of major global banks from observation to active infrastructure development. Banks occupy a uniquely strategic position in any tokenisation ecosystem: they are simultaneously originators of the assets to be tokenised, custodians of the client relationships through which tokens are distributed, and holders of the trust relationships, the KYC/AML infrastructure, and the regulated status, and the counterparty credit assessments that underpin the entire system.

JPMorgan's Kinexys platform, formerly known as Onyx, is one of the most visible institutional blockchain initiatives in settlement and payments. Public reporting indicates that the platform has been used for significant transaction volumes and for experiments in intraday and cross-currency repo, but any precise claims about cumulative value or the first occurrence of a specific transaction should be tied to a dated source.

HSBC's Orion platform facilitated the first cross-border repo transaction involving a natively issued digital bond on a public blockchain, spanning three jurisdictions, Switzerland, Singapore and Japan. The transaction involved borrowing tokenised Japanese yen against a JPY-denominated digital bond, with the full cycle, including redemption, principal and interest payments, executing entirely on-chain. This was not a simulation or a pilot; it was a live transaction with real economic consequences.

Société Générale's FORGE platform has run full bond issuances on Ethereum. Goldman Sachs and BNY Mellon have launched tokenised money market funds. In Asia, OCBC Bank became the first Singapore institution to deploy tokenisation in full production for corporate bonds in January 2025, not as an experiment but as part of its daily treasury and capital markets operations.

Asset Managers: Scale and Standardisation

“The participation of large asset managers has been an important signal of institutional credibility and growing scale. BlackRock is among the world's largest asset managers, and its public commentary on tokenisation reflects a broader shift from experimentation toward operational implementation.

BUIDL illustrates institutional adoption of tokenised funds, but any claim about it being the world's largest tokenised fund should be qualified by date, methodology and source, given that market rankings and AUM data evolve quickly

Franklin Templeton's FOBXX fund, running on the Stellar and Polygon blockchains, similarly tokenises US Treasury investments, allowing institutional and accredited retail investors to hold shares as blockchain tokens. The fund was among the first to demonstrate that regulatory-compliant tokenised investment products could operate within existing US securities law, which is a crucial proof point for the broader market.

“With BlackRock managing \$14 trillion in assets, building tokenisation infrastructure, and describing it as inevitable, this development is not hype. This is strategic commitment from the world's most influential asset manager.”

Regulatory Innovation Hubs

Governments and central banks have played a more active and constructive role in the development of tokenisation than is sometimes appreciated. Rather than simply regulating from a distance, several leading regulatory bodies have embedded themselves directly in the development process through supervised experimentation.

The Monetary Authority of Singapore's Project Guardian, the most comprehensive central bank-led tokenisation experiment globally, has involved multiple cohorts of financial institutions testing different aspects of tokenised asset markets, from fixed income and foreign exchange to asset management and trade finance. The initiative has produced detailed findings that are shaping regulatory frameworks across Asia and beyond. MAS has explicitly committed to building interoperable tokenisation infrastructure that complements, rather than displaces, traditional market infrastructure.

The Hong Kong Monetary Authority's Project Ensemble has similarly brought together financial institutions to test tokenised asset settlement using wholesale CBDCs, with trade finance explicitly identified as a priority use case. The European Investment Bank has issued digital bonds on Ethereum. The BIS Innovation Hub has run projects across multiple central bank partnerships exploring CBDCs, tokenised collateral and cross-border payment infrastructure.

In the United States, regulatory engagement around tokenisation has become more active in recent years, with the SEC and congressional committees showing growing interest in the issue. The broader policy environment has also been influenced by stablecoin discussions and by the search for clearer rules governing tokenised financial instruments, although the timing, scope and significance of specific hearings and legislative proposals should be stated only when directly sourced.

The Emerging Ecosystem: Technology and Platform Providers

Alongside the banks and asset managers, a rich ecosystem of enabling infrastructure has developed. Securitise has emerged as a leading regulated tokenisation platform in the US market, partnering with BlackRock on BUIDL and enabling multiple other institutional token issuances. Centrifuge and Maple Finance have built decentralised credit infrastructure that enables businesses to tokenise invoices and receivables for direct DeFi investment. Ondo Finance has established a dominant position in tokenised US treasuries, controlling approximately 60% market share through its Global Markets platform by early 2026.

In the trade finance space specifically, the GSBN (Global Shipping Business Network) has implemented electronic bill of lading tokenisation with Ant Digital Technologies, enabling trade settlements using stablecoins, CBDCs or tokenised deposits, explicitly designed to bridge the trade finance gap. XDC Network has built blockchain infrastructure specifically optimised for trade finance applications, with partnerships across multiple development finance contexts.

The McKinsey Projection and Market Trajectory

Looking at the aggregate trajectory, the projections from major consultancies and financial institutions are consistently bullish. McKinsey forecasts \$4–5 trillion in digital securities issuance by 2030. Ripple and BCG project tokenised RWAs expanding from approximately \$0.6 trillion in 2025 to \$18.9 trillion by 2033. The 2026 RWA.xyz market data shows the market on a trajectory consistent with these projections, with growth accelerating rather than decelerating as institutional infrastructure matures.

\$30T+

Projected RWA tokenisation market by
2034 (various estimates)

200+

Active institutional RWA token initiatives
globally

CHAPTER 7

Risks, Challenges and the Road Not Yet Taken

Intellectual honesty demands that the genuine risks and challenges of trade finance tokenisation receive the same analytical rigour as its opportunities. Technological optimism that ignores structural barriers is not analysis; it is advocacy. And the barriers to tokenisation scaling to its full potential are real, substantial and in some cases genuinely difficult to resolve.

Technology Risks

Smart Contract Vulnerabilities

Smart contracts, once deployed, cannot be modified, or at best can be modified only through governance mechanisms that themselves carry risks. A bug in a smart contract that manages hundreds of millions of dollars of investor funds can have catastrophic, irreversible consequences. The history of the broader crypto ecosystem includes numerous examples of smart contract exploits resulting in the permanent loss of significant capital.

For institutional-grade financial products, this risk is partially mitigated through rigorous pre-deployment auditing, formal verification of contract logic, and phased deployment with limited initial exposure. But it cannot be eliminated entirely. Institutional participants must develop smart contract risk assessment capabilities, either in-house or through specialist third parties, that are currently rare within traditional financial institutions.

Oracle Risk

Many tokenised trade finance applications depend on oracles, which are external data feeds that provide on-chain smart contracts with real-world information (shipping confirmations, customs clearances, payment verifications). The oracle is necessarily a point of trust in an otherwise trustless system: if the oracle is compromised, delayed or incorrect, the smart contract may execute incorrectly, with potentially serious consequences. The Financial Stability Board has explicitly identified oracle risk as a material concern for tokenised financial systems.

Interoperability Fragmentation

The tokenisation ecosystem has developed in a fragmented manner, with different institutions building on different blockchain platforms and using different technical standards. A token issued on the Ethereum network cannot natively interact with a token on a Corda network or a Hyperledger Fabric-based platform. This fragmentation limits liquidity; if secondary markets are divided across multiple incompatible platforms, none of them achieve sufficient depth to be genuinely useful. Resolving interoperability is technically achievable; cross-chain bridges, standardised APIs and the development of universal token standards (such as the IEEE blockchain interoperability standard) all contribute but require coordination across competing institutions with competing commercial interests.

Legal and Regulatory Risks

Jurisdictional Uncertainty

The legal status of tokenised assets varies significantly across jurisdictions. In some jurisdictions – Singapore, Switzerland, the UK, and Abu Dhabi – regulatory frameworks are sufficiently developed to provide reasonable certainty about how tokenised securities are treated under existing law. In many others, including major trading nations in Asia, Africa and Latin America, the legal and regulatory treatment is ambiguous or explicitly unsettled.

This creates a practical problem for cross-border tokenised trade finance. A token representing a receivable originated in Bangladesh, held by an investor in Germany, on a blockchain governed by Singapore law, whose legal framework applies in the event of default? Which jurisdiction's insolvency laws govern the SPV? Can the token holder enforce the underlying receivable in the jurisdiction where the obligor is located? These are not hypothetical questions; they are the questions that legal counsel must be able to answer definitively before institutional investors will commit capital at scale.

True Sale and Bankruptcy Remoteness

The legal foundation of any securitisation structure, tokenised or otherwise, is the true sale analysis: the determination that assets transferred from the originator to the SPV are genuinely off the originator's balance sheet and beyond the reach of the originator's creditors in insolvency. In well-developed legal systems with mature ABS markets, this analysis is settled by precedent. In many developing market jurisdictions targeted by trade finance tokenisation, it is not, and establishing it may require litigation that could take years.

Classification Uncertainty

The classification of tokenised instruments: are they securities or derivatives? notes? derivative instruments? payment instruments? They have direct implications for the regulatory obligations that apply to their issuance, trading and custody. A token classified as a security in the United States requires registration or exemption under the Securities Act; the same token classified as a payment instrument would fall under a completely different regulatory framework. Until major jurisdictions have definitively resolved these classification questions, institutional participants face the risk of retrospective regulatory reclassification.

Market Structure Risks

Concentration and Systemic Risk

As tokenisation scales, new systemic risks may arise. If large proportions of trade finance assets globally migrate to a small number of blockchain platforms, those platforms become systemically important infrastructure, as implied. The Financial Stability Board's October 2024 report on the financial stability implications of tokenisation explicitly flagged this concern, noting that tokenisation could create new forms of interconnectedness and contagion that existing regulatory frameworks are not designed to address.

Liquidity Illusion

There is a risk that tokenisation creates an appearance of liquidity without its substance. A token may be technically transferable 24 hours a day, 7 days a week, but if there are no willing buyers in a market stress scenario, its liquidity is illusory. The history of structured finance contains sobering precedents: mortgage-backed securities that appeared liquid in normal conditions proved impossible to sell in 2007-08 without catastrophic price impacts. Tokenisation does not solve the fundamental problem of liquidity in stressed markets; it can only improve the infrastructure through which liquidity is expressed when it genuinely exists.

“Analytically, tokenisation can make a liquid asset more efficiently liquid by improving transferability, settlement speed and market access. It cannot, however, turn a fundamentally illiquid asset into a liquid one simply by placing it on a blockchain.”

Operational Risks

Custody of digital assets presents challenges that are fundamentally different from conventional asset custody. Private keys that control access to blockchain wallets must be secured against theft, loss and misappropriation, a technical challenge that has defeated numerous crypto-native entities but is increasingly being addressed by institutional-grade custody providers. Legal clarity on the obligations of regulated custodians who hold tokenised assets is developing but not yet uniform across jurisdictions.

Cybersecurity represents a related and material concern. Blockchain infrastructure is not inherently immune to cyberattacks: while the distributed nature of public blockchains makes the ledger itself extremely difficult to compromise, the surrounding infrastructure, exchanges, custodians, oracle providers, and tokenisation platforms present attack surfaces that require the same level of defensive investment as any systemically important financial infrastructure.

What Must Be True

For tokenised trade finance to scale beyond pilots and niche issuance, several preconditions must be satisfied simultaneously. The first is legal enforceability: token holders must have clear, defensible rights to the underlying asset, and those rights must be recognised consistently across relevant jurisdictions. Without this foundation, tokenisation remains a technical wrapper rather than a functional market structure.

The second is standardised data. Trade finance depends on documents, counterparties, shipment data, payment flows and compliance records, all of which must be machine-readable and harmonised if automation is to work at scale. Interoperability between platforms, data models and document standards is therefore not optional; it is the condition that allows tokenised assets to move across systems without losing integrity or legal meaning.

The third requirement is trusted oracles and a robust compliance infrastructure. Smart contracts can only be as reliable as the external data they consume, whether that data comes

from customs systems, logistics providers, payment networks or sanctions-screening tools. If the oracle layer is weak, delayed or manipulable, the automation layer becomes a source of error rather than efficiency. This is why tokenised finance requires not just code, but governed interfaces between on-chain logic and off-chain reality.

The fourth is interoperable rails. Tokenised trade finance will not scale if every institution builds a closed system that cannot communicate with other institutions. Liquidity, investor access and operational efficiency all depend on networks that can connect across jurisdictions and platforms, rather than fragment into isolated digital silos. Interoperability is therefore a market-structure issue, not just a technology preference.

The fifth is genuine investor demand. Even the most elegant structure will not scale if investors do not see a credible risk-return profile, transparent servicing, and reliable exit options. Tokenised trade finance must compete with conventional fixed-income, private-credit and short-duration instruments on both yield and operational convenience. Demand will depend on whether tokenised assets can offer something materially better than existing alternatives, rather than simply something more novel.

The sixth is reliable custody. Institutional adoption requires secure key management, transfer controls, recovery procedures and integration with existing portfolio infrastructure. Custody is not a peripheral service in tokenised markets; it is part of the market's core plumbing, because control of the asset is inseparable from control of the keys that govern it. Without institution-grade custody, tokenised assets will remain difficult to hold, transfer and govern at scale.

Taken together, these conditions define the boundary between tokenisation as a compelling concept and tokenisation as a scalable market infrastructure. The analytical value of this section is that it makes the paper more realistic: it shows that tokenisation is not a single solution but a system that only works when legal, technical, commercial and custodial prerequisites are aligned.

CHAPTER 8

Regulatory Frontiers

The regulatory dimension of tokenisation is simultaneously one of its greatest current constraints and one of its most significant opportunities. Well-designed regulatory frameworks will not slow tokenisation – they will accelerate it by providing the legal certainty that institutional capital requires before committing at scale. The jurisdictions that get this right will attract the infrastructure, talent and liquidity that define the financial centres of the next generation.

The Leading Regulatory Frameworks

Singapore

Singapore has established the most comprehensive and progressive regulatory framework for tokenised financial markets among major financial centres. The Monetary Authority of Singapore has explicitly endorsed tokenisation as a strategic priority, provided regulatory accommodation for experimental tokenisation structures through its capital markets services licensing regime, and embedded itself directly in the development process through Project Guardian. MAS has signalled its intention to develop interoperability standards for tokenised asset markets, recognising that fragmented national systems will limit the efficiency gains that make tokenisation valuable.

Switzerland

Switzerland's DLT Act, which came into force in 2021, created a dedicated legal category for 'DLT securities', allowing securities to be issued natively on distributed ledgers with full legal force, without the need to maintain a parallel paper or conventional book-entry record. This was a landmark legislative achievement that resolved the fundamental legal uncertainty about the status of tokenised instruments. As a result, Switzerland has attracted numerous tokenisation platforms and issuers.

The European Union

In the EU, the most relevant framework for tokenised securities is the DLT Pilot Regime, which allows regulated market infrastructures to test the trading and settlement of tokenised financial instruments under supervisory conditions. MiCA applies more broadly to crypto-assets, but tokenised securities are generally addressed through securities-law frameworks rather than MiCA. The European Investment Bank's digital bond issues have provided practical experience, but they should be described as market examples rather than as direct regulatory drivers.

United Kingdom

In the UK, the relevant framework is the Financial Services and Markets Act 2023, together with the Digital Securities Sandbox, which provides a regulated environment for testing and operating tokenised securities. This framework is more precise than referring to a 'Financial Markets and Services Act 2023', which is incorrect.

United States

The US regulatory landscape has been more complex. The SEC's approach to crypto-assets, often characterised as enforcement-first, created uncertainty that slowed institutional tokenisation development. However, 2025-2026 marked a significant shift. The SEC's Crypto Task Force, under Commissioner Hester Peirce, articulated a more accommodating framework for tokenised securities. The GENIUS Act's stablecoin framework provided settlement infrastructure certainty. Congressional hearings demonstrated bipartisan interest in enhancing US competitiveness in tokenised markets. The landscape is still evolving, but the trajectory is clearly toward greater regulatory clarity.

The Developing Market Imperative

For tokenised trade finance to fulfil its potential in addressing the trade finance gap, regulatory frameworks must develop not only in financial centres like Singapore, London and New York, but also in the developing market jurisdictions where the gap is most acute. This situation presents a different set of challenges.

Many developing market jurisdictions lack the legal infrastructure, the case law, the specialist regulators, and the legal profession expertise to rapidly adapt existing frameworks for digital assets. Building this capacity takes time and requires international technical assistance. Multilateral development banks, the World Bank, ADB, and AfDB have roles to play in supporting regulatory capacity building that are at least as important as their roles as investors in tokenised instruments.

There is also a governance question: regulatory frameworks for tokenised assets that affect multiple jurisdictions must ultimately be coordinated internationally, or else fragmentation will undermine the efficiency gains that tokenisation promises. The BIS, the Financial Stability Board and the International Organization of Securities Commissions are all engaged in developing international standards, but the pace of regulatory standard-setting inevitably lags the pace of market development.

What Good Regulation Looks Like

Based on the experience of leading jurisdictions and the analysis of what market participants actually need, a set of regulatory design principles emerges:

1. Legal clarity on the status of tokens: the governing law framework must definitively establish what rights token holders have, against whom, and how they are enforceable.
2. Bankruptcy remoteness: the law must clearly support true sale treatment for assets transferred to SPVs, including in insolvency scenarios.
3. Technology-neutral frameworks: regulation should define the outcomes required (investor protection, market integrity, systemic stability) without prescribing the specific technology through which those outcomes are achieved.
4. Interoperability standards: regulators should actively support the development of technical standards that enable different tokenisation platforms to communicate and transact with each other.
5. Proportionate sandboxes have proved more effective than either blanket prohibition or unlimited permissiveness by allowing supervised experimentation regimes that enable innovation to proceed within defined limits and inform permanent frameworks.
6. International coordination: the cross-border nature of tokenised trade finance requires international regulatory coordination that national frameworks alone cannot provide.

CHAPTER 9

The Future State: Convergence, Scale and Impact

Projection is always uncertain, but grounded analysis of the forces at work permits a view of what the tokenised trade finance market might look like a decade from now and what the implications would be for the global economy, for financial inclusion, and for the actors who position themselves well in advance.

The Convergence of Traditional and Tokenised Markets

The binary framing of 'traditional finance' versus 'DeFi', once ubiquitous in discussions of blockchain in finance, is already obsolete among sophisticated practitioners. The trajectory is clearly toward convergence: a market structure in which traditional financial infrastructure and token-enabled infrastructure coexist, interconnect and ultimately merge into a unified system that combines the regulatory robustness and institutional trust of traditional finance with the efficiency, transparency and programmability of blockchain-based systems.

This convergence is visible in specific institutions. BlackRock is simultaneously the world's largest traditional asset manager and the operator of the world's largest tokenised fund. JPMorgan runs one of the most sophisticated traditional investment banks in the world and operates blockchain infrastructure that processes trillions in settlement transactions. OCBC Bank offers conventional banking services and manages tokenised bond portfolios as part of its routine treasury operations. These developments do not contradict one another; rather, they represent the natural evolution of institutions as they adapt their infrastructure to new capabilities.

The future market structure will likely feature public utility infrastructure, blockchain networks with broad participation and standardised protocols, sitting underneath permissioned access layers that enforce compliance, identity and regulatory requirements. Assets will flow between these layers based on their characteristics and the needs of the transaction. Investors will access tokenised and conventional instruments through unified interfaces, unaware of the underlying infrastructure differences and indifferent to them.

The Projected Market Trajectory

Year	Estimated On-Chain RWA Market	Key Drivers
2023	\$5 billion	Early institutional pilots, limited asset classes
2024	\$10–12 billion	BlackRock BUIDL, Franklin Templeton, regulatory clarity
2025	\$24 billion	Institutional scaling and tokenised treasuries boom
2026 (current)	\$32+ billion	Mainstream adoption, deep secondary markets forming
2030 (projected)	\$2–9 trillion	Regulatory convergence, standardisation, global scale
2034 (projected)	\$30+ trillion	Full convergence of traditional and tokenised markets

Sources: RWA.xyz market data; Ripple/BCG projection; Standard Chartered/Synpulse market sizing analysis; McKinsey Global Institute. Note: Estimates vary significantly across sources due to differing definitions and methodologies.

The Impact on Global Trade Finance

If the optimistic projections for the broader tokenisation market are realised, what does the outcome imply for trade finance specifically? Market sizing analysis suggests that tokenised trade finance assets could constitute up to 16% of the total tokenised asset market by 2034, a figure that would represent several trillion dollars. Given that the total addressable market for tokenised trade finance (including the trade finance gap) is estimated at \$14 trillion, this figure would represent a significant but incomplete penetration of the opportunity.

Even partial adoption of tokenised trade finance could have meaningful implications for the trade finance gap. If it helps channel additional funding to underserved businesses, it could improve access to capital for SMEs in developing markets, deep-tier supply chain participants and firms operating in corridors where conventional trade finance has become more limited. The scale of any gap reduction, however, should be presented as a scenario rather than a forecast, since outcomes will depend on regulation, investor appetite, legal enforceability and market infrastructure.

The implications for global trade volumes are significant. Trade finance is not a passive response to trade flows, it is a prerequisite for many of them. Businesses that lack affordable financing simply do not trade, or they trade at significantly reduced volumes. Narrowing the financing gap directly expands global trade, with knock-on effects for employment, income and development outcomes in the economies that participate.

Programmable CBDCs and the Next Frontier

Looking beyond the immediate horizon, the integration of programmable central bank digital currencies (CBDCs) with tokenised trade finance infrastructure represents the most powerful potential accelerant for the market. CBDCs, digital versions of fiat currency issued by central banks, combine the stability and finality of central bank money with the programmability of blockchain-based digital assets.

A programmable CBDC can be issued to an anchor buyer, who passes it to a tier-one supplier as a conditional payment, who uses it as collateral to obtain financing from their bank, who passes it further down the supply chain as payment to tier-two and tier-three suppliers. The CBDC carries with it the programming that specifies when and how it converts to free-use cash, for example, upon confirmation of delivery or upon completion of a quality inspection. The entire supply chain finance flow is automated, transparent and settled in central bank money.

Multiple central banks are in advanced CBDC exploration. Hong Kong, Singapore, China, the European Central Bank and the UK are all progressing wholesale CBDC programmes that explicitly include trade finance applications. When these programmes mature and interconnect, a process that may take until the late 2020s, the combination of tokenised trade assets and programmable settlement currency will represent a genuinely transformative infrastructure for global trade.

Financial Inclusion at Scale

The ultimate test of tokenisation's transformative power is not whether it benefits institutional investors or global banks; it will clearly do both. The test is whether it reaches the businesses and communities that conventional finance has failed.

There are credible reasons for cautious optimism. Tokenisation reduces the fixed costs of origination, distribution and administration that currently make small-ticket trade finance uneconomic for banks. It enables the credit of strong anchor buyers to flow through supply chains to suppliers that are too small to access financing independently. It creates investment vehicles that can channel global institutional capital, from pension funds, sovereign wealth funds and insurance companies, into the real economic activity of developing markets.

But optimism must be conditional. Technology does not automatically redistribute power. If the same incumbent institutions that currently dominate trade finance control tokenisation infrastructure, they may simply automate existing patterns of exclusion more efficiently. Realising the inclusive potential of tokenisation requires deliberate choices, by regulators, by multilateral institutions, by investors and by the banks and platforms that build tokenisation infrastructure, to prioritise access over concentration.

“Tokenisation does not automatically democratise finance. But it creates conditions under which democratisation is possible for the first time, if the actors with power choose to use it that way.”

CHAPTER 10

Actions at the Inflection Point

We are at an inflection point. The technology is proven. The institutional infrastructure is being built. The regulatory frameworks are crystallising. The market exists and is growing. But the full potential of tokenised trade finance will not be realised automatically; it requires deliberate choices and actions from each category of stakeholder in the ecosystem.

For Institutional Investors

The case for building expertise in tokenised assets is now more urgent than ever. As the market grows from tens of billions to hundreds of billions to trillions, first-mover investors will be able to negotiate better terms, access a wider range of opportunities and build the operational infrastructure on their own timelines rather than under competitive pressure.

- **Invest in education:** Build internal expertise in DLT technology, smart contract risk, digital asset custody and tokenised market structures. This is a specialised knowledge domain that cannot be acquired overnight.
- **Participate in pilots:** Engage with regulated tokenisation pilots and experimentation programmes. Practical operational experience is worth more than theoretical analysis as preparation for scaled participation.
- **Develop custody solutions:** Resolve the operational question of how tokenised assets will be held, transferred and managed within existing portfolio management infrastructure.
- **Engage with regulators:** Participate in regulatory consultations and sandboxes. Regulators who hear from sophisticated institutional investors are better positioned to develop frameworks that enable responsible market development.
- **Allocate deliberately:** Begin allocating a defined, modest portion of portfolios to tokenised trade finance assets, enough to build genuine operational experience, not so much that early-stage market risks are existential.

For Banks and Financial Institutions

Banks are the indispensable intermediaries in tokenised trade finance: originators of the underlying assets, custodians of client relationships, holders of regulatory licences and providers of the trust infrastructure that underpins the system. They cannot disintermediate themselves, but institutions that move faster can displace them.

- **Build originate-to-distribute infrastructure:** Develop the platforms, legal structures and operational processes that enable trade finance assets to be tokenised and distributed to capital markets investors systematically.

- **Join or build interoperable networks:** Proprietary, closed tokenisation platforms limit liquidity and investor access. The network effects of interoperable, industry-wide infrastructure are likely to overwhelm proprietary alternatives over time.
- **Invest in KYC/AML for digital assets:** The compliance infrastructure for tokenised markets is different from conventional compliance, but equally essential. Banks that build this capability create a sustainable competitive advantage.
- **Partner with technology providers:** The technology stack for tokenised finance requires capabilities that most banks do not have in-house. Strategic partnerships with specialist tokenisation platforms, smart contract developers and DLT infrastructure providers are necessary.
- **Engage proactively with regulators:** The regulatory frameworks that govern tokenised markets are still being written. Banks that engage constructively with regulators, sharing operational experience, identifying genuine risks, and proposing solutions, will shape frameworks that enable their businesses rather than constrain them.

For Governments and Regulators

The regulatory choices made in the next three to five years will determine whether tokenised trade finance scales to its potential or remains a niche market constrained by legal uncertainty. The financial stakes are significant, but the implications extend beyond just money. The regulatory jurisdictions that enable responsible tokenisation at scale will attract substantial investment, talent and infrastructure.

- **Provide legal clarity on token status:** The single most important action any regulator can take is definitively establishing the legal status of tokenised instruments under their jurisdiction's law, what rights they confer, how they are enforced and how they are treated in insolvency.
- **Support true sale frameworks:** Legal infrastructure that clearly enables assets to be transferred out of originator balance sheets with bankruptcy-remote treatment is foundational to the securitisation of any kind, tokenised or conventional.
- **Invest in regulatory capacity:** Regulators who understand blockchain technology, smart contracts and DeFi protocols are more effective than those who do not. Investment in regulatory expertise is an investment in regulatory quality.
- **Coordinate internationally:** Tokenised trade finance is inherently cross-border. Unilateral national frameworks create fragmentation. Active engagement in BIS, FSB and IOSCO standards development is a necessary complement to national regulation.
- **Support developing market inclusion:** Advanced economy regulatory frameworks that do not consider the needs of developing market participants risk accelerating, rather than reducing, global financial exclusion. Development finance institutions should actively fund regulatory capacity building in emerging markets.

For Multilateral Institutions

The World Bank, International Finance Corporation, Asian Development Bank, African Development Bank and their peers have unique roles to play – as investors in tokenised instruments, as guarantors that bridge the creditworthiness gap for developing market borrowers, as technical assistance providers for regulatory capacity building, and as conveners of the international coordination that national actors alone cannot achieve.

- Invest in tokenised trade finance instruments: Multilateral participation signals creditworthiness and attracts private capital. First-loss guarantees and partial credit enhancements from development finance institutions can bridge the risk gap for institutional investors considering developing market exposure.
- Provide technical assistance: Regulatory capacity building, legal framework development and transaction advisory for tokenisation structures in developing markets are high-leverage interventions.
- Convene standard-setting: The international coordination required for tokenised trade finance to scale requires conveners with the legitimacy and neutrality to bring competing institutions to agreement. Multilaterals are uniquely positioned to play this role.

The Window Is Narrowing

In markets that are undergoing structural transformation, the window for advantaged positioning is finite. Early movers gain experience, relationships, regulatory influence and operational infrastructure that latecomers must pay a premium to replicate. The tokenised trade finance market is moving from early experimentation to early-stage mainstream adoption. The institutions that act decisively in the next two to three years will define the market structure for the following decade.

“In transformative markets, timing is not everything, but it is close. The infrastructure being built today will determine who participates, on what terms, in the tokenised trade finance market of 2030.”

Glossary

Asset-Backed Security (ABS)

A financial instrument whose value and income payments are derived from and collateralised by a pool of underlying assets.

Blockchain

A distributed ledger that is updated in groups of transactions called blocks, chained sequentially via cryptography to form an immutable record.

Central Bank Digital Currency (CBDC)

The digital form of a country's fiat currency, issued and backed by the central bank, with the potential to be programmable via smart contracts.

Decentralised Finance (DeFi)

Financial services and products built on public blockchain networks, operating without centralised intermediaries such as banks or brokerages.

Deep-Tier Supply Chain Finance

Financing extended not just to direct (tier-one) suppliers of a large anchor buyer but to suppliers further down the supply chain through token-enabled credit transfer.

Distributed Ledger Technology (DLT)

A database that is consensually shared and synchronised across multiple sites, institutions or geographies, with no central administrator.

ERC-20

The technical standard for fungible tokens on the Ethereum blockchain, defining a common set of rules for token behaviour and enabling interoperability.

Ethereum

A decentralised blockchain platform that introduced programmable smart contracts, enabling the creation of diverse tokenised instruments.

Fungible Token

A digital token where each unit is identical to every other unit of the same class and can be exchanged on a one-for-one basis.

Initial Token Offering (ITO)

The primary market issuance of tokens representing financial assets, functionally analogous to an IPO or bond issuance in traditional capital markets.

KYC/AML

Know Your Customer / Anti-Money Laundering are regulatory compliance requirements for financial institutions to verify client identities and detect financial crime.

Letter of Credit (LC)

A financial instrument issued by a bank guaranteeing a buyer's payment to a seller, contingent on the presentation of specified documents evidencing shipment of goods.

Missing Middle

Middle market enterprises (MMEs), businesses too large for microfinance but too opaque for investment-grade trade finance, representing the largest underserved segment in global trade finance.

Non-Fungible Token (NFT)

A unique digital token that represents a specific asset with distinct attributes, as opposed to fungible tokens which are interchangeable.

Oracle

A service that connects blockchain smart contracts to external, real-world data (shipping confirmations, payment records, market prices), enabling contracts to respond to off-chain events.

Real-World Asset (RWA)

A type of digital token that derives its value from and represents ownership of an asset that exists outside the blockchain, whether physical, financial or otherwise.

Security Token Offering (STO)

A regulated token issuance in which the tokens are classified and treated as securities under applicable law, subject to investor protection requirements.

Smart Contract

Self-executing computer code deployed on a blockchain that automatically enforces and executes the terms of an agreement when predefined conditions are met.

Special Purpose Vehicle (SPV)

A legal entity created specifically to hold assets in a securitisation or tokenisation structure, providing bankruptcy remoteness from the originator.

Stablecoin

A digital asset designed to maintain a stable value relative to a reference asset, typically a fiat currency, used as a medium of exchange and settlement in digital asset markets.

Tokenisation

The process of creating a digital representation of a real or traditional asset on a distributed ledger technology platform in the form of a transferable, programmable token.

Trade Finance

Financial products and instruments (letters of credit, trade receivables, and supply chain finance facilities) that facilitate international trade by bridging payment timing mismatches between buyers and sellers.

Trade Finance Gap

The trade finance gap is the difference between the amount of trade finance that businesses globally request and the amount they actually receive, which the Asian Development Bank estimated at \$2.5 trillion in 2023.

Tranching

The division of a pool of assets into segments (tranches) with different risk, return and priority of payment profiles, enabling different investor segments to access the same underlying assets.

True Sale

A legal determination that assets transferred from an originator to an SPV are genuinely removed from the originator's balance sheet and beyond the reach of its creditors in insolvency.